

Confidence Intervals

Lecture 10

Topics in this lecture

- Central Limit Theorem
- Large-sample confidence interval for a population mean

Central Limit Theorem

- If we draw a large enough sample from the population, then the distribution of the *sample mean* is approximately *normal*, no matter what population the sample was drawn from
- Let $X_1, X_2, \dots, X_n$ be a simple random sample drawn from a population with mean μ and variance σ^2 ,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

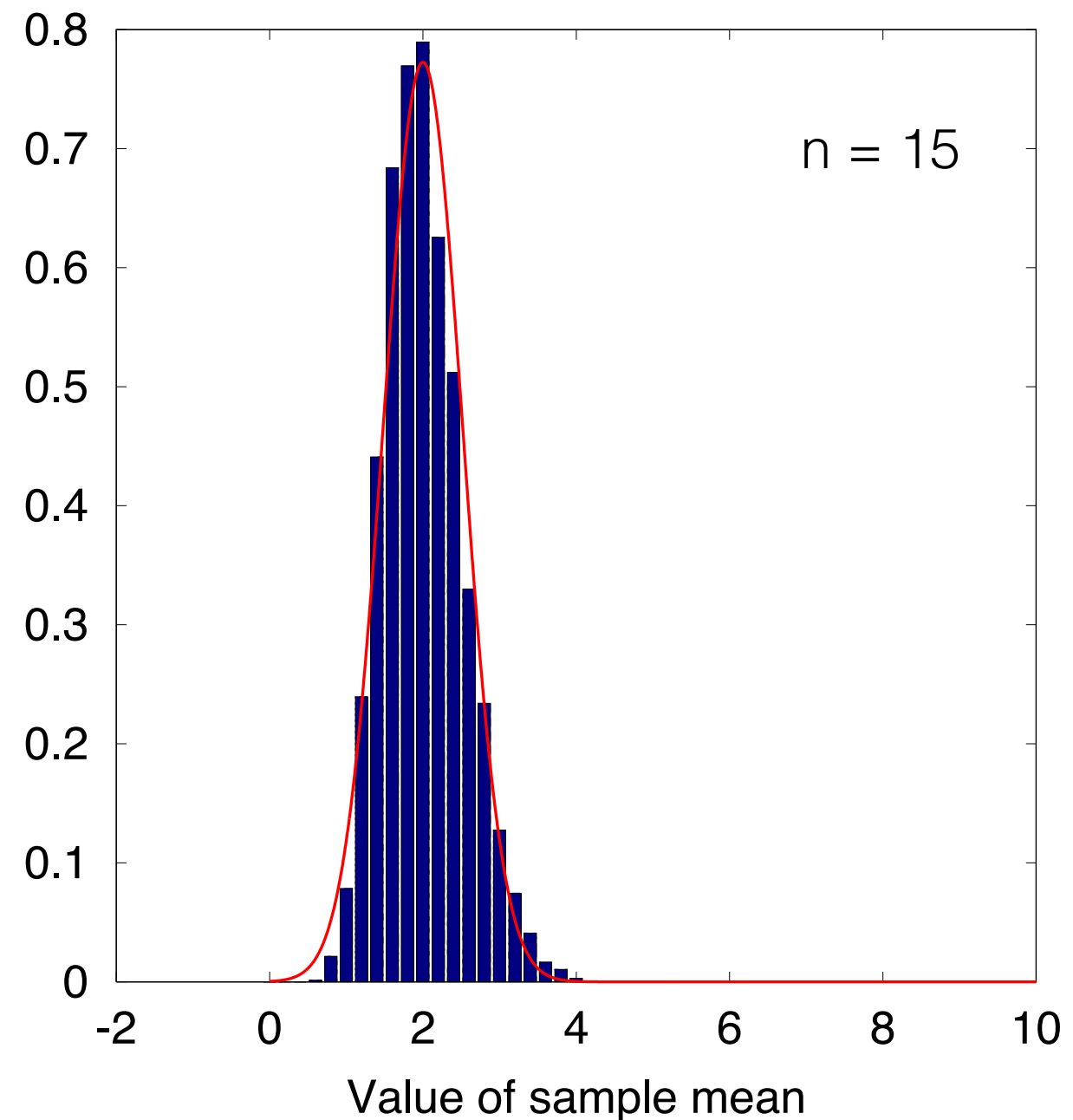
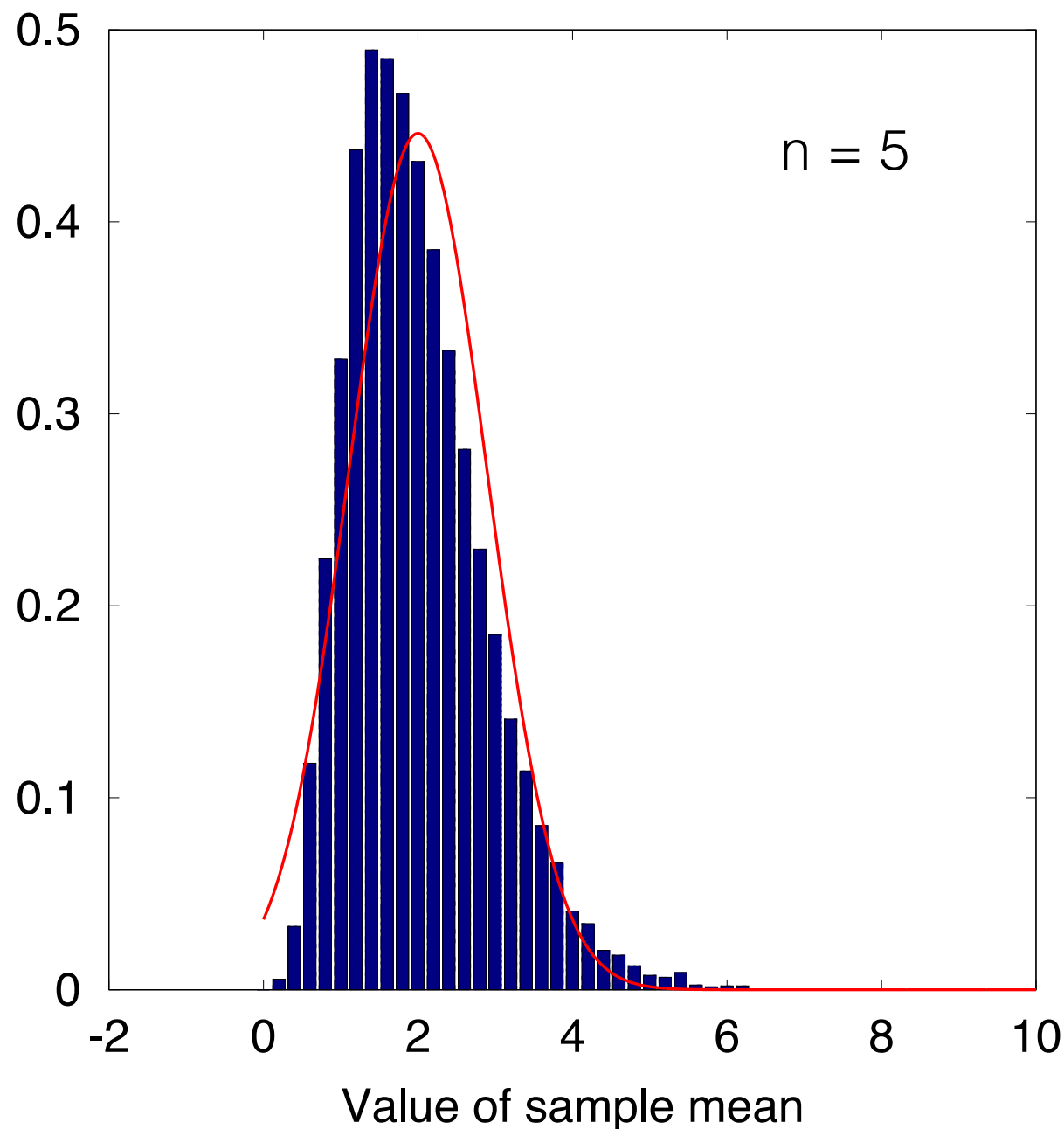
$$\bar{X} \sim \mathcal{N} \left(\mu, \frac{\sigma^2}{n} \right)$$

if n is large enough ($n \geq 30$)


$$S_n = \sum_{i=1}^n X_i$$

$$S_n \sim \mathcal{N} (n\mu, n\sigma^2)$$

Distribution of Sample Mean



Distribution of sample mean. Samples are drawn from a population with $\text{Exp}(0.5)$

Example 1 *z-table*

- In a large school, the mean weight of the students is 53.4 kg, and the standard deviation is 6 kg. A random sample of 50 students is drawn. What is the probability that the average weight of these students will be greater than 55 kg?

$$\mu = 53.4, \sigma = 6$$

$$P(\bar{X} > 55)$$

$$\bar{X} \sim N\left(53.4, \frac{6^2}{50}\right)$$

$$P(\bar{X} > 55) = 1 - P(\bar{X} \leq 55)$$

$$P(\bar{X} > 55) = 1 - P\left(Z \leq \frac{55 - 53.4}{\sqrt{\frac{6^2}{50}}}\right)$$

$$= 1 - \Phi(1.89)$$

$$= 1 - 0.9706 = 0.0294$$

Confidence Intervals

- Sample mean is only an estimator of the true mean. It is not exactly equal to the true mean.
- We want to assess the level of *confidence* that the true mean falls within a certain interval

Confidence Interval for Population Mean

- Let $X_1, X_2, \dots, X_n$ be a large ($n \geq 30$) random sample from a population with mean μ and standard deviation σ .

- We know that

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

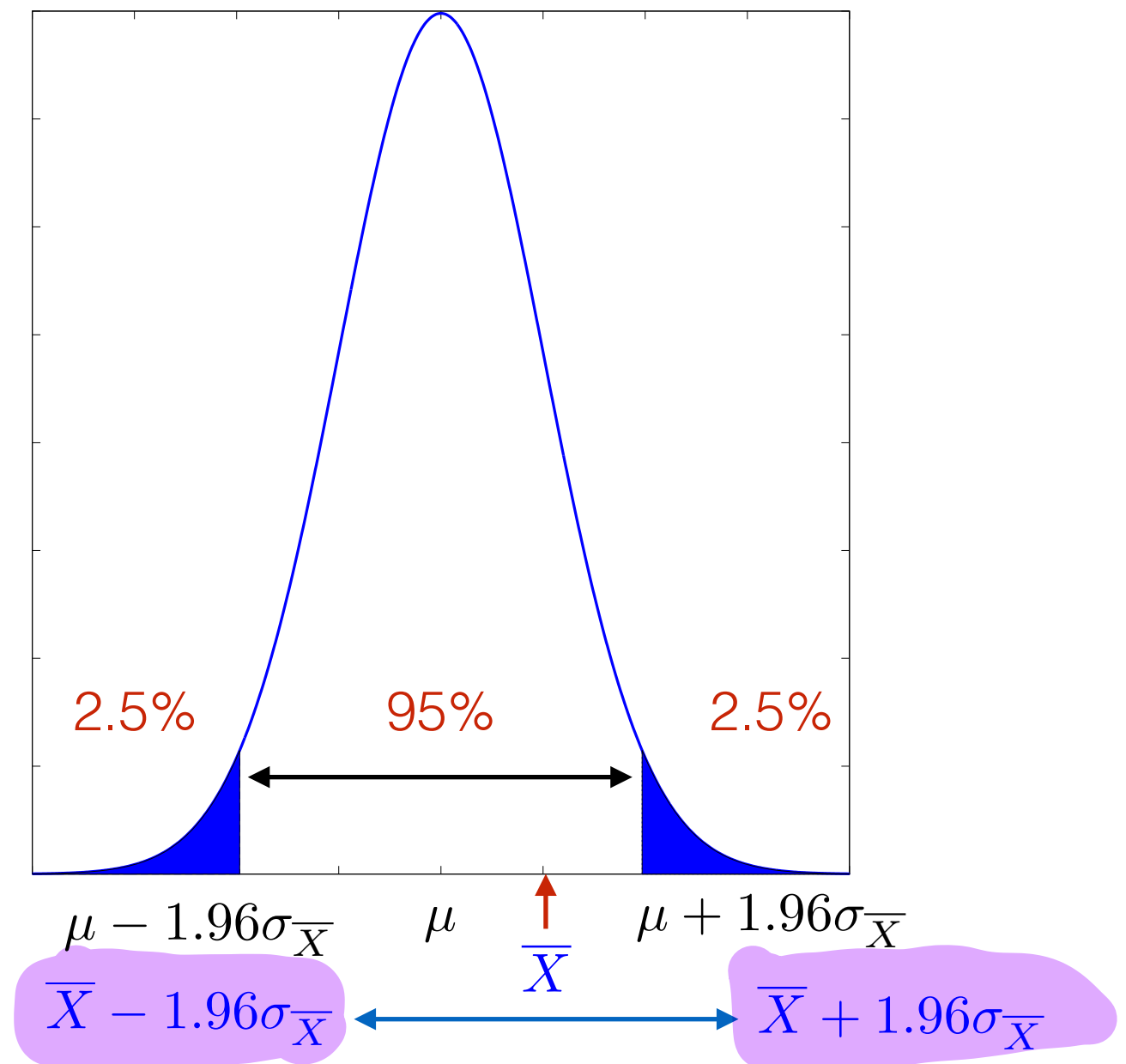
$$\sigma_{\bar{X}} = \sigma / \sqrt{n}$$

- The middle 95% is within

$$\mu \pm 1.96\sigma_{\bar{X}}$$

95% confidence interval

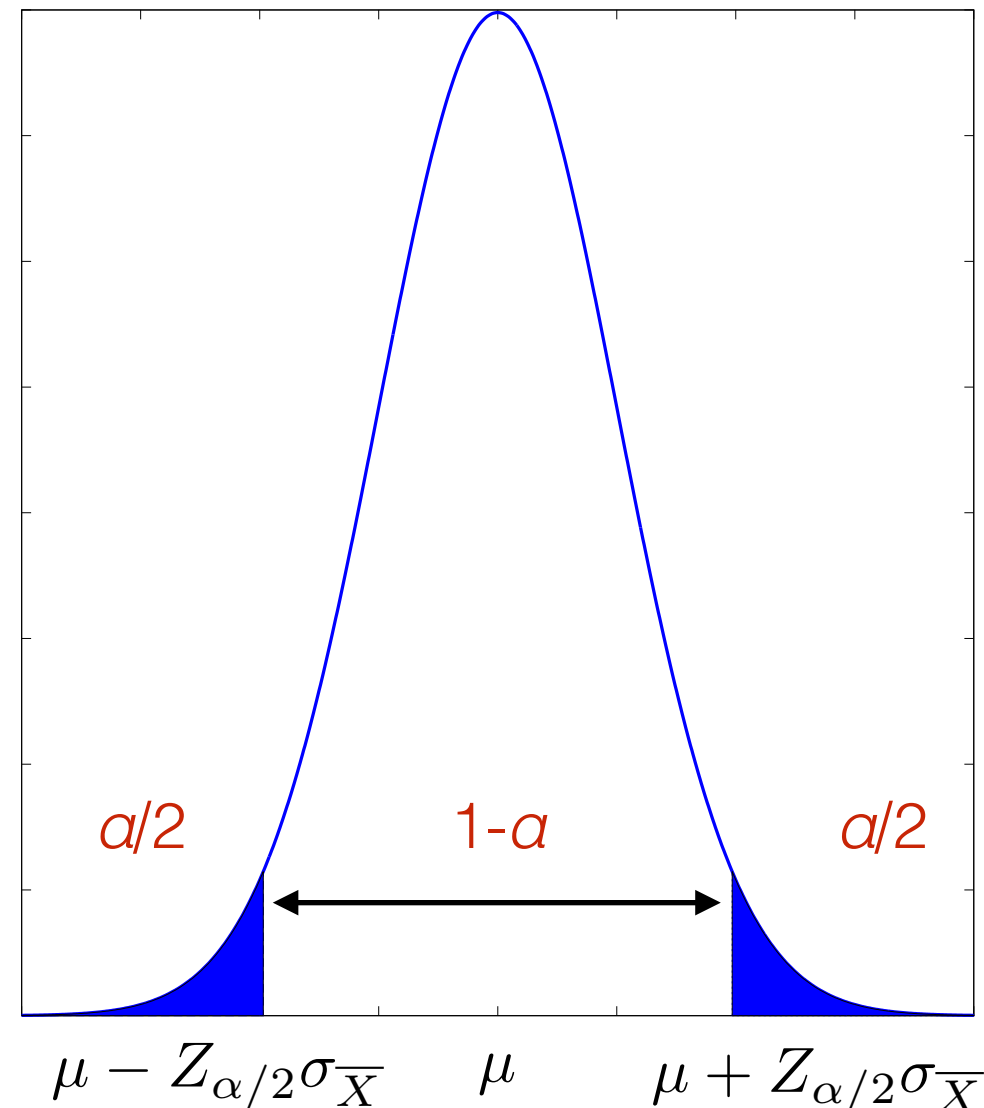
$$\bar{X} \pm 1.96\sigma_{\bar{X}}$$



Confidence Interval

- Let $X_1, X_2, \dots, X_n$ be a large ($n \geq 30$) random sample from a population with mean μ and standard deviation σ .
- Let α be a number in $[0,1]$
- Let $z_{\alpha/2}$ be the z-score that cuts off an area of $\alpha/2$ in the right-tail
- 100(1- α)% confidence interval for μ is

$$\bar{X} \pm Z_{\alpha/2} \sigma_{\bar{X}}$$



Large-Sample Confidence Intervals for a Population Mean

- Let $X_1, X_2, \dots, X_n$ be a large ($n \geq 30$) random sample from a population with mean μ and standard deviation σ , so that the sample mean $\bar{X}$ is approximately normal. Then a level $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{X} \pm Z_{\alpha/2} \sigma_{\bar{X}}$$

where $Z_{\alpha/2}$ is the z-score that cuts off an area of $\alpha/2$ on the right-tail

$$\sigma_{\bar{X}} = \sigma / \sqrt{n}$$

When the value of σ is unknown, it can be replaced with sample standard deviation s .

Values of $Z_{\alpha/2}$ at different C.I.

C.I. Level	$Z_{\alpha/2}$
68%	1
90%	1.645
95%	1.96
99%	2.58
99.7%	3

Example 2

- The sample mean and standard deviation for the fill weights of 100 boxes are $\bar{X} = 12.05$ and $s = 0.1$. Find

- 68% confidence interval

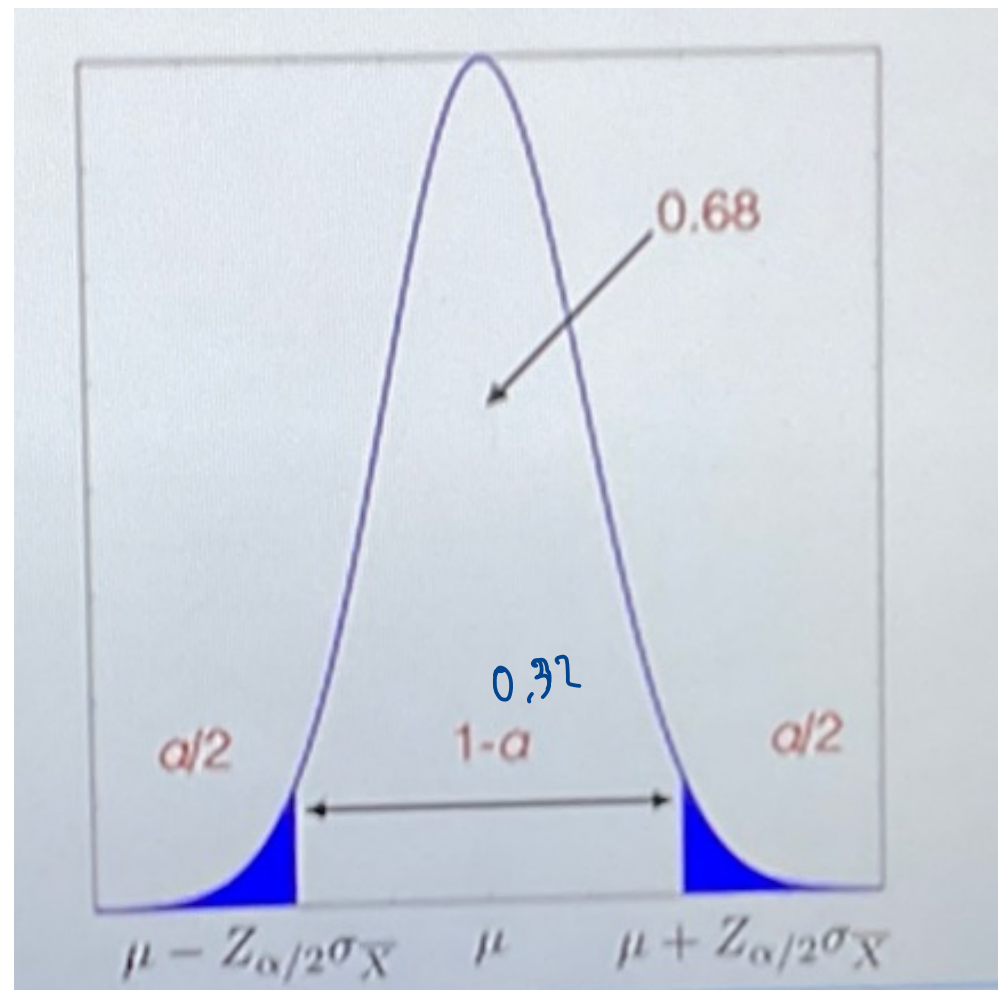
$$\alpha = 0.32, \alpha/2 = 0.16$$

$$Z_{\alpha/2} = 1 \quad (1 - 0.8413 \approx 0.16)$$

$$\bar{X} \pm Z_{\alpha/2} s \sqrt{n}$$

$$12.05 \pm 1 \cdot \frac{0.1}{\sqrt{100}}$$

$$68\% \text{ CI is } (12.04, 12.06)$$



Example 2

- The sample mean and standard deviation for the fill weights of 100 boxes are $\bar{X} = 12.05$ and $s = 0.1$. Find

- 85% confidence interval

$$\alpha = 0.15, \alpha/2 = 0.075$$

$$Z_{\alpha/2} = 1.44 \quad (\pm 0.9251 \approx 0.075)$$

$$\bar{X} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$12.05 \pm 1.44 \times \frac{0.1}{\sqrt{100}}$$

$$85\% \text{ is } (12.036, 12.064)$$

Example 3

- A sample of 50 students has the following sample mean and standard deviation for heights, $\bar{X} = 164.2$ cm and $s = 12.5$ cm. Find
 - 68% confidence interval
 - 95% confidence interval
 - 99% confidence interval

Example 4

- Find the levels of the confidence intervals that have the following values of $Z_{\alpha/2}$
 - $Z_{\alpha/2} = 1.28$
 - $Z_{\alpha/2} = 2.17$
 - $Z_{\alpha/2} = 3.28$

Determining Sample Size Needed for a C.I. with Specified Width

- Sometimes the C.I. is too wide to be useful, and we want to make it narrower. To do this, the sample size must be increased.
- Ex: The sample standard deviation of weights from 100 boxes was $s = 0.1$ oz. How many boxes must be sampled to obtain a 99% confidence interval of width ± 0.012 oz

$$\alpha = 0.01, \frac{\alpha}{2} = 0.005$$

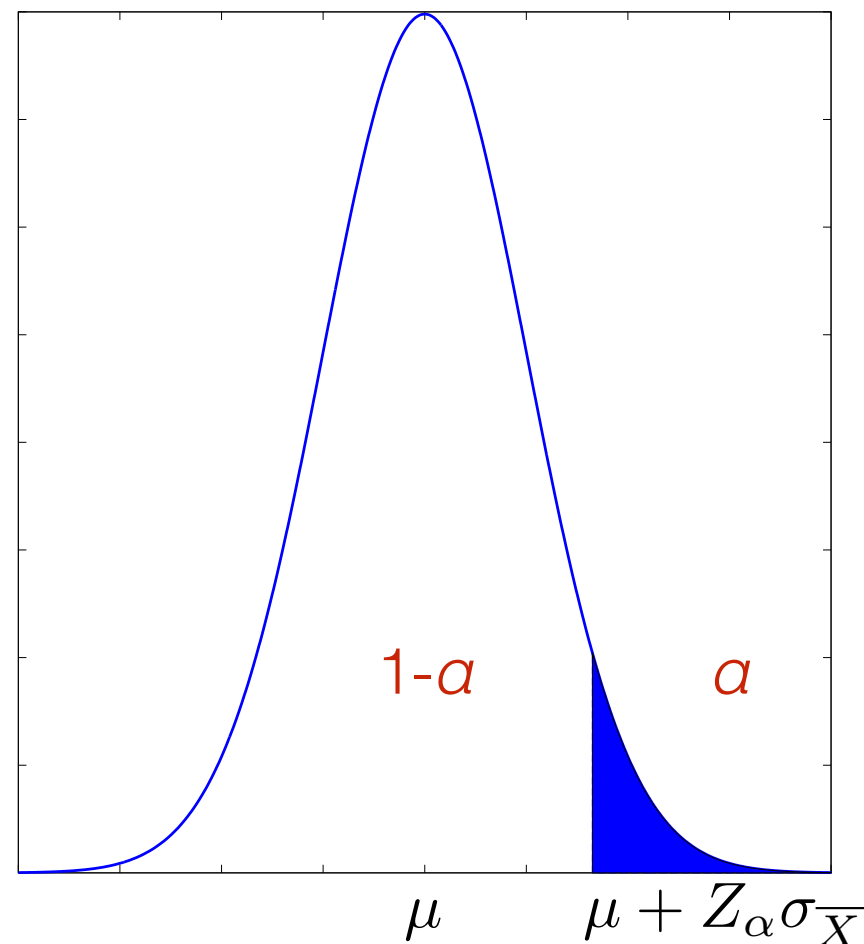
$$Z_{0.005} = 2.58$$

$$0.012 = 2.58 \times \frac{0.1}{\sqrt{n}}$$

$$n \approx 463$$

One-sided C.I.

- The confidence intervals discussed so far have been two-sided in that they specify both a lower and upper confidence bounds
- In some applications, we are only interested only in one of these bounds



One-sided C.I. (2)

- Let $X_1, X_2, \dots, X_n$ be a large ($n \geq 30$) random sample from a population with mean μ and standard deviation σ , so that the sample mean $\bar{X}$ is approximately normal. Then a level $100(1 - \alpha)\%$ lower confidence interval bound for μ is

$$\bar{X} - Z_{\alpha} \sigma_{\bar{X}}$$

and level $100(1 - \alpha)\%$ upper confidence bound or μ is

$$\bar{X} + Z_{\alpha} \sigma_{\bar{X}}$$

When the value of σ is unknown, it can be replaced with sample standard deviation s .

Values of Z_α at different C.I. (One-sided)

C.I. Level	Z_α
90%	1.28
95%	1.645
99%	2.33

Example 5

- A sample of 50 students has the following sample mean and standard deviation for heights, $\bar{x} = 164.2$ cm and $s = 12.5$ cm. Find
- one-sided 95% lower confidence bound

$$\bar{x} - z_{\alpha} \sigma_{\bar{x}}$$

$$\alpha = 0.05$$

$$z_{0.05} = 1.645$$

$$\bar{x} - z_{\alpha} \sigma_{\bar{x}}$$

$$164.2 - 1.645 \times \frac{12.5}{\sqrt{50}} = 161.29$$

Example 5

- A sample of 50 students has the following sample mean and standard deviation for heights, $\bar{x} = 164.2$ cm and $s = 12.5$ cm. Find
 - one-sided 99% upper confidence bound

$$\bar{x} + z_{\alpha} \frac{s}{\sqrt{n}} \quad \alpha = 0.01$$

$$164.2 + 2.33 \times \frac{12.5}{\sqrt{50}} = 168.32$$